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Short-term benefits of osteopathic manipulative treatment for chronic low back pain: systematic review and meta-analysis

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Abstract

Context: Chronic low back pain (CLBP) is common and often managed within multimodal, nonpharmacologic pathways. Clarifying sham-controlled effects of osteopathic manipulative treatment (OMT) can guide early clinical decisions.

Objectives: This study aims to estimate the short-term (≤ 6 weeks) effects of OMT for CLBP, emphasizing sham-controlled comparisons.

Methods: We preregistered a systematic review and meta-analysis (PROSPERO [Prospective Register of Systematic Reviews] CRD420251154852) and followed PRISMA 2020 (Preferred Reporting Items for Systematic reviews and Meta-Analyses 2020). Randomized trials enrolling adults with CLBP compared OMT with sham/attenuated OMT, usual care, or wait-list. The co-primary outcomes were pain (0–10; mean difference [MD]) and function (Oswestry Disability Index [ODI] or Roland–Morris Disability Questionnaire [RMDQ]; standardized mean difference [SMD]). Analyses were prespecified at end-of-treatment (≤ 6 weeks); the single crossover trial contributed pre-crossover data only. Searches covered MEDLINE (Medical Literature Analysis and Retrieval System Online), Embase (EMBASE [Excerpta Medica dataBASE]), CENTRAL (Cochrane Central Register of Controlled Trials), CINAHL (Cumulative Index of Nursing and Allied Health), and ClinicalTrials.gov through October 2025. Random-effects models (DerSimonian–Laird with Hartung–Knapp)

summarized effects with I^2 and prediction intervals (PIs); risk of bias (RoB 2) and certainty (GRADE [Grading of Recommendations, Assessment, Development, and Evaluation]) were assessed.

Results: Seven trials ($n \approx 378$) met the inclusion criteria. Across five trials, OMT reduced pain vs. comparators: MD -1.32 on 0–10 scales (95 % confidence interval [CI], -1.71 to -0.93 ; $I^2=6$ %; 95 % PI, -2.05 to -0.59). In sham-controlled analyses of function limited to ODI/RMDQ ($k=2$), OMT improved disability: SMD 1.75 (95 % CI, 1.37 to 2.13 ; $I^2=0$ %). Adverse events were infrequently reported; where monitored, no serious adverse events were attributed to OMT. Certainty by GRADE was moderate for short-term pain and for sham-controlled function.

Conclusions: At ≤ 6 weeks, OMT reduces pain and improves function vs. sham without observed serious harms, supporting its use as a low-risk adjunct within guideline-concordant, nonopioid CLBP care.

Keywords: chronic low back pain; manual therapy; meta-analysis; osteopathic manipulative treatment; physical function; randomized controlled trials

Introduction

Chronic low back pain (CLBP) remains a leading cause of disability and healthcare spending, with roughly 1 in 10 adults reporting persistent pain that limits activity and work participation [1–4]. Contemporary guidelines emphasize multimodal, nonpharmacologic care and caution against early pharmacologic escalation given modest average benefits and dose-related harms [5–9]. Within this landscape, clinicians and patients need decision-ready estimates for manual therapies that can be integrated safely alongside exercise, education, and psychologically informed care. National physician guidelines based on systematic reviews specifically recommend nonpharmacological treatments – including education, graded exercise, and psychologically informed care – as first-line approaches for patients with CLBP.

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Osteopathic manipulative treatment (OMT) is a physician-delivered, hands-on intervention that individualizes a broad repertoire of techniques – including but not limited to soft tissue, myofascial release, muscle energy, articulatory, counterstrain, high-velocity low-amplitude (HVLA) thrust, visceral, and cranial techniques – within a biopsychosocial framework [10–12]. Proposed mechanisms include neuromodulatory and biomechanical effects with circulatory/lymphatic contributions, consistent with osteopathic principles of structure–function interdependence. However, despite widespread use, estimates of OMT’s specific effects – particularly those isolating expectancy and performance influences via sham procedures over short, clinically relevant windows – remain fragmented across small, randomized trials [11, 13]. Focusing on short-term outcomes is clinically salient because treatment decisions and patient expectations are often formed during the initial 2–6 weeks of care [5–7].

To address this gap, we conducted a preregistered systematic review and meta-analysis of randomized trials in adults with CLBP, emphasizing sham-controlled comparisons and restricting analyses to end-of-treatment (≤ 6 weeks) to reflect early, clinically actionable decision points. We prespecified pain on scales of 0–10 (mean difference [MD]) and physical function on the Oswestry Disability Index (ODI) or Roland–Morris Disability Questionnaire (RMDQ; standardized mean difference [SMD]), synthesized effects with random-effects models utilizing Hartung–Knapp adjustments, summarized heterogeneity with I^2 and PIs, assessed risk of bias with RoB 2, and graded certainty with Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) [13–18].

Our objective was to deliver precise, clinically interpretable short-term estimates of OMT’s benefits and harms to inform guideline-concordant care and to guide future comparative-effectiveness trials.

Methods

Protocol and reporting

We conducted a systematic review and meta-analysis of randomized controlled trials (RCTs), preregistered in PROSPERO (Prospective Register Of Systematic Reviews, CRD420251154852) and reported per PRISMA 2020 (Preferred Reporting Items for Systematic reviews and Meta-Analyses 2020) [14]. The protocol specified the population, comparators, outcomes, time window, and analytic plan.

Eligibility criteria

We included RCTs enrolling adults (≥ 18 years) with CLBP (duration ≥ 3 months) that compared OMT (any technique combination delivered by trained clinicians) with sham/attenuated OMT, usual care, or wait-list [13]. Co-primary outcomes were pain on 0–10 scales (higher=worse) and physical function on the ODI or RMDQ. We prespecified the end-of-treatment window as ≤ 6 weeks (or the nearest assessment within ≤ 6 weeks). The single crossover trial contributed pre-crossover data only to avoid carryover [19]. Trials focusing on postoperative, specific radicular, or non-lumbar conditions were excluded [12].

Interventions and comparators

OMT was defined as physician-delivered manual care utilizing a broad repertoire of techniques, including but not limited to soft tissue, myofascial release, muscle energy, articulatory, counterstrain, HVLA thrust, visceral, and cranial techniques. “Sham/attenuated OMT” referred to credible touch/positioning procedures designed to mimic OMT without delivering therapeutic force or intent [13]. When trials had multiple comparators, we selected the comparator aligned with the synthesis (e.g., sham for sham-restricted analyses) and combined same-category arms to avoid double-counting.

Information sources and search strategy

We searched MEDLINE (Medical Literature Analysis and Retrieval System Online [Ovid]), Embase (EMBASE [Excerpta Medica dataBASE]), Cochrane CENTRAL (Cochrane Central Register of Controlled Trials), Cumulative Index of Nursing and Allied Health (CINAHL), and ClinicalTrials.gov from inception through October 2025, supplemented by forward and backward citation searches [14]. Searches were conducted in October 2025 by M.S. and J.K. The core concepts searched across all databases were: (1) CLBP (e.g., “low back pain,” lumbago, combined with chronic\$); (2) OMT and named techniques (e.g., osteopathic manipulation\$, OMT, HVLA, high-velocity low-amplitude, muscle energy, counterstrain, myofascial release, articulatory, Still technique, lymphatic pump); and (3) randomized/controlled trial filter with human and adult limits. Database-specific syntax (Medical Subject Headings [MeSH] in MEDLINE, Emtree in Embase, CINAHL subject headings, and Cochrane Central Register of Controlled Trials [CENTRAL] thesaurus terms) was combined with free-text synonyms utilizing Boolean

operators. No language or publication-status limits were imposed at the search stage. The search strategies are provided in Appendix 1.

Study selection

Two reviewers (M.S., J.K.) independently screened titles/abstracts and full texts in duplicate, with consensus resolution of disagreements [14]. Reasons for exclusion at full text were recorded and are summarized in the PRISMA flow diagram.

Data extraction and management

Utilizing piloted forms, two reviewers (M.S., J.K.) independently extracted study design, setting, participant characteristics, intervention/comparator details (including dose/frequency), outcome instruments and time points, group *N*s, means, and dispersion (standard deviation [SD] or standard error [SE]) [13]. For multi-arm trials, we combined arms when appropriate. When SDs were not reported, we derived them from SE ($SD = SE \times \sqrt{n}$) or back-calculated from 95 % CIs, *t* statistics, or *p* values per standard methods. We contacted authors for clarifications when needed. Function effects were oriented so that positive values favored OMT; standardized effects utilized Hedges *g* to correct small-sample bias.

Risk of bias and certainty of evidence

Risk of bias was assessed at the outcome level with RoB 2 across randomization, deviations from intended interventions, missing outcome data, measurement of outcomes, and selection of reported results [17]. Certainty of evidence was graded with GRADE considering risk of bias, inconsistency, indirectness, imprecision, and publication bias, with explicit footnotes in the Summary of Findings table [17].

Effect measures and synthesis

For pain, we computed MDs on 0–10 scales; for function (ODI/RMDQ), we computed SMDs (Hedges *g*) [13]. We utilized DerSimonian–Laird random-effects models with Hartung–Knapp adjustments [15, 16]. Between-study heterogeneity was summarized with I^2 and τ^2 , and we reported 95 % PI to communicate dispersion of true effects [13]. With fewer than 10 trials per synthesis, small-study effect

tests were considered underpowered and not utilized for inference.

Handling multiplicity and unit of analysis

When studies reported multiple eligible short-term time points, we utilized the assessment closest to end-of-treatment within ≤ 6 weeks [13]. If multiple pain or function instruments were reported at the same time, we followed a prespecified hierarchy (ODI/RMDQ prioritized for function; 0–10 Numeric Rating Scale [NRS]/Visual Analog Scale [VAS] prioritized for pain) to avoid multiplicity. For crossovers, only pre-crossover (T2) data were included [19]. Clustered data were adjusted if clustering was ignored and sufficient information was provided; otherwise, such studies were excluded from quantitative synthesis [12].

Missing data

We utilized available-case analysis [13]. When change scores and baseline data were both reported, we prioritized change scores; when only endpoint data were available, we utilized endpoints. If SDs were missing and could not be derived, studies were narratively described but excluded from the corresponding meta-analysis.

Sensitivity and exploratory analyses

A prespecified sensitivity pooled function across instruments (ODI/RMDQ plus Patient-Reported Outcomes Measurement Information System – Physical Function [PROMIS-PF] from the crossover trial, pre-crossover) was utilized to probe the robustness of sham-controlled signals; given high heterogeneity in the mixed-instrument pool, these estimates were interpreted as sensitivity only [13, 19, 20]. We also explored influence diagnostics (leave-one-out) when *k* permitted.

Software and reproducibility

Analyses were performed in R (R v4.4.1) utilizing the metafor (v4.6-0) and meta packages (v6.5-0) [13]. Data and code to reproduce the analyses are available as Appendix 2; the Summary of Findings tables include GRADE footnotes and decision thresholds [17].

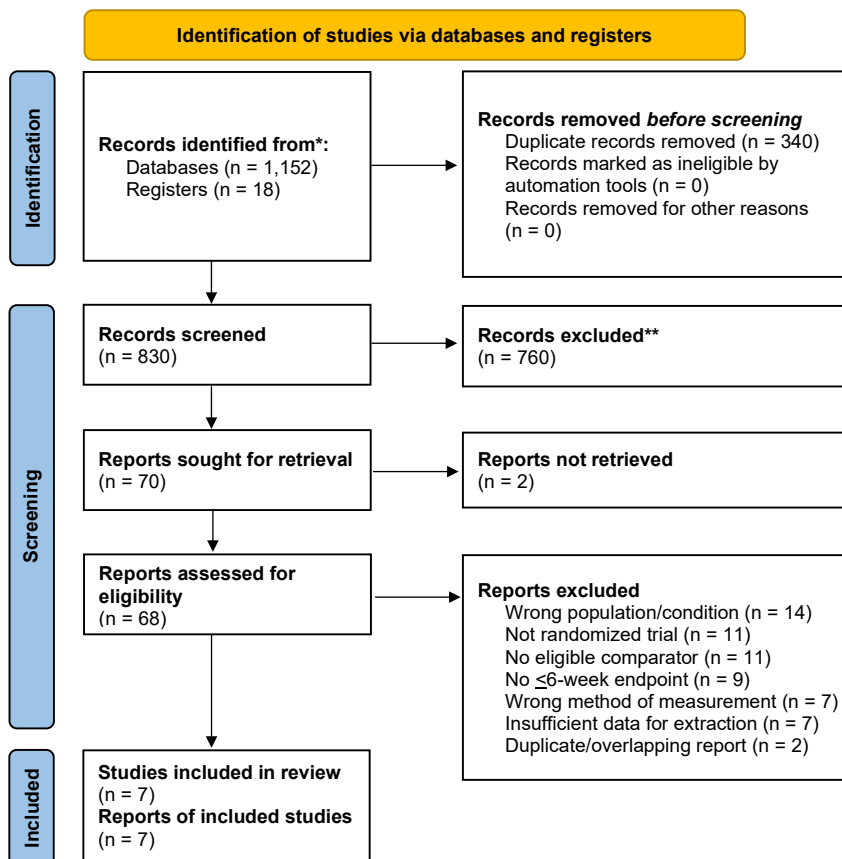


Figure 1: A PRISMA 2020 flow diagram depicting the literature search and study selection process. Searches covered MEDLINE (Ovid), Embase, Cochrane CENTRAL, CINAHL, and ClinicalTrials.gov through October 2025. Seven RCTs met the inclusion criteria. The reasons for exclusion at full-text review are presented in the diagram. CENTRAL, Cochrane Central Register of Controlled Trials; CINAHL, Cumulative Index of Nursing and Allied Health; MEDLINE, Medical Literature Analysis and Retrieval System Online; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

Results

Study selection and characteristics

Seven randomized trials ($n \approx 378$) met the inclusion criteria, enrolling ambulatory adults with nonspecific CLBP and comparing individualized OMT protocols with sham/attenuated OMT, usual care, or wait-list [21–26]. The study selection process is depicted in Figure 1 (PRISMA flow diagram). Across trials, OMT drew from a concise physician-delivered repertoire (soft tissue, myofascial release, muscle energy, articular, counterstrain, and selective HVLA), and the single crossover trial contributed pre-crossover data only to avoid carryover. Most studies reported outcomes at end-of-treatment within ≤ 6 weeks, aligning with our prespecified window and supporting like-with-like pooling [12].

Pain

Across five trials, OMT reduced pain vs. comparators: pooled MD -1.32 on 0–10 scales (95 % CI, -1.71 to -0.93), with low between-study heterogeneity ($I^2=6\%$) and a 95 % PI, -2.05 to -0.59 [21–26]. Forest plots for pain outcomes are

presented in Figure 2. Interpreted clinically, this magnitude aligns with thresholds patients perceived as noticeable over short treatment horizons [27, 28]. Directionality was consistent across contributing trials, and leave-one-out influence analyses did not materially alter the point estimate where k permitted [12].

Function (sham-controlled)

In analyses restricted to sham-controlled trials reporting ODI/RMDQ ($k=2$), OMT improved disability with a large standardized effect: SMD 1.75 (95 % CI, 1.37 to 2.13 ; $I^2=0\%$), with both trials pointing in the same direction [23, 26]. Forest plots for function outcomes are presented in Figure 3. This pattern suggests a robust functional signal when identical instruments are utilized and expectancy/performance influences are minimized by sham procedures [12].

Sensitivity (mixed instruments)

A prespecified sensitivity analysis pooled ODI/RMDQ with PROMIS-PF from the crossover trial (pre-crossover) and

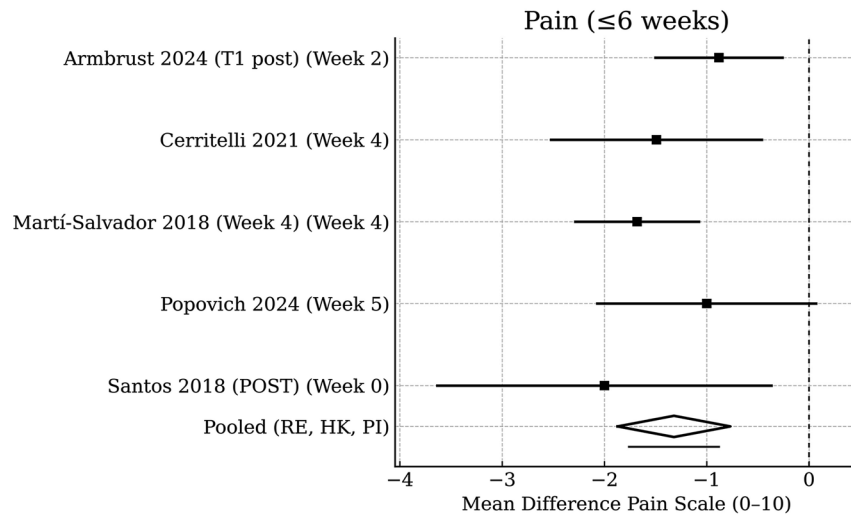


Figure 2: Forest plot of pain outcomes (MD on 0–10 scales) for OMT vs. comparators in adults with CLBP. Analysis includes five RCTs ($n \approx 378$). The pooled MD was -1.32 (95 % CI, -1.71 to -0.93 ; $I^2=6\%$; 95 % PI, -2.05 to -0.59). Effect estimates and 95 % CIs are shown for each trial; the diamond represents the pooled estimate from a random-effects model (DerSimonian–Laird with Hartung–Knapp adjustment). Negative values favor OMT. CI, confidence interval; MD, mean difference; OMT, osteopathic manipulative treatment; PI, prediction interval.

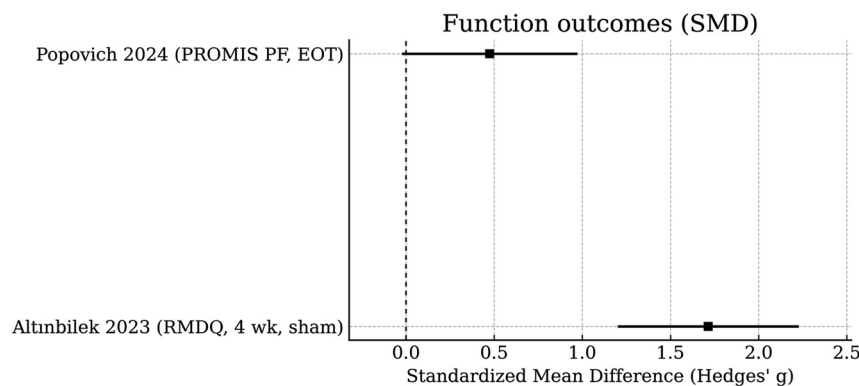


Figure 3: Forest plot of function outcomes (standardized mean difference) for OMT vs. sham in adults with CLBP. Analysis is restricted to sham-controlled trials reporting the ODI or RMDQ ($k=2$). The pooled SMD was 1.75 (95 % CI, 1.37 to 2.13 ; $I^2=0\%$). Effect estimates and 95 % CIs are shown for each trial; the diamond represents the pooled estimate from a random-effects model (DerSimonian–Laird with Hartung–Knapp adjustment). Positive values favor OMT. CI, confidence interval; ODI, Oswestry disability Index; OMT, osteopathic manipulative treatment; RMDQ, Roland–Morris disability Questionnaire; SMD, standardized mean difference.

remained directionally positive but highly heterogeneous: SMD 1.20 (95 % CI, 0.08 to 2.32; $I^2 \approx 93\%$; Hartung–Knapp 95 % CI, -1.82 to 4.21 ; 95 % PI, -12.93 to 15.33) [13, 19, 23, 26]. Given between-instrument and design heterogeneity, we treat these estimates as sensitivity-only and do not utilize them for decision-making.

Harms

Adverse events were infrequently reported; where monitored, no serious adverse events were attributed to OMT delivered by trained clinicians [21–26]. Harms ascertainment methods varied, which limits rare-event detection [12].

Risk of bias and certainty

Most outcomes were rated as having some concerns on RoB 2 – typically for deviations from intended interventions or

selective reporting – with no trial being at high risk [17]. Certainty by GRADE was moderate for short-term pain and for sham-controlled function, limited primarily by small k per synthesis and by between-instrument and design heterogeneity in the mixed-instrument sensitivity pool [13, 18]. Detailed RoB 2 domain ratings are provided in Appendix 2, and the full GRADE Summary of Findings table with footnotes also appears in Appendix 2.

Overall evidence profile

Taken together, the evidence indicates that, in ambulatory adults with CLBP, OMT reduces short-term pain (MD -1.32 on 0–10 scales; 95 % CI, -1.71 to -0.93 ; 95 % PI, -2.05 to -0.59 ; moderate certainty) and improves function vs. sham when identical instruments are utilized (SMD 1.75; 95 % CI, 1.37 to 2.13; moderate certainty), with no serious adverse events observed in trials that monitored harms [18, 21–26]. Interpretation and clinical context are discussed below [12].

Discussion

Summary of findings

Across seven randomized trials in ambulatory adults with CLBP, OMT produced short-term improvements in pain and disability compared with sham/attenuated comparators or usual care, with no serious adverse events reported when harms were monitored [21–26]. The direction and consistency of effects – with low heterogeneity for pain and a large, concordant signal for function when identical instruments were utilized – support a specific therapeutic contribution of OMT over expectancy and performance influences minimized by sham procedures [13]. Clinically, the short-horizon analgesia aligns with the thresholds patients perceive as noticeable and may facilitate near-term activity restoration within multimodal care [27].

Relation to prior evidence and guidelines

These sham-focused estimates extend previous manual-therapy syntheses by narrowing to randomized, short-term, instrument-matched comparisons and by applying decision-oriented analyses (Hartung–Knapp, PIs) [13]. The pattern of benefit is concordant with contemporary guidelines that prioritize nonpharmacologic, multimodal pathways and caution against early pharmacologic escalation given modest average benefits and dose-related harms [5–9]. Within that framework, OMT is positioned as a physician-delivered, low-risk adjunct that can be integrated alongside exercise, education, and psychologically informed care to support function without medication intensification [5–7, 10, 12]. For comparison, several randomized trials of OMT in CLBP have evaluated outcomes beyond the 6-week window examined here; these longer-term studies provide important context for interpreting the durability of the short-term effects reported in this review and should be described in future iterations of this work [29–31].

Strengths and limitations

Strengths include prespecified restriction to end-of-treatment (≤ 6 weeks), the use of pre-crossover data only to avoid carry-over, and analytic choices that communicate both average effects and the dispersion of likely true effects across settings (PIs) [13, 15, 16, 19]. The primary limitation is the small number of trials per synthesis – particularly for function when restricted to identical instruments – along with frequent RoB 2

“some concerns” ratings and variable harms ascertainment that limits rare-event detection [13, 17]. Generalizability is limited because trials focused on ambulatory adults with nonspecific CLBP rather than postoperative or radicular populations. Heterogeneity rose when function instruments were mixed (ODI/RMDQ with PROMIS-PF), underscoring the value of instrument consistency for decision-making [13, 25].

Clinical implications

For ambulatory adults with CLBP, these findings support offering OMT as a pragmatic adjunct within guideline-concordant care – particularly when the goal is patient-noticeable pain relief and early functional gains while avoiding pharmacologic escalation. In practice, OMT should be paired with graded activity, education, and psychologically informed care to reinforce early gains and minimize medication escalation [5–7, 10, 28].

Research priorities

Future trials should standardize outcome sets (e.g., ODI/RMDQ with complementary PROMIS domains and validated minimal important differences), prespecify OMT dosing and technique bundles to enable dose–response analyses, extend follow-up to assess durability to ≥ 12 –24 weeks, and include comparative-effectiveness designs vs. other nonpharmacologic therapies [11, 13, 18, 20]. Consistent harms terminology and prospective capture are needed, as are mechanistic substudies that probe neuromodulatory and circulatory/lymphatic pathways hypothesized to mediate benefit.

Conclusions

In ambulatory adults with CLBP, OMT yields short-term, patient-noticeable analgesia and meaningful functional improvement compared with sham, with no serious adverse events reported in monitored trials. These findings support offering OMT as a low-risk adjunct within guideline-concordant, multimodal care. Future trials should standardize outcome sets, specify dosing/technique bundles, and assess durability to ≥ 12 –24 weeks.

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Conflict of interest: None declared.

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Data availability: All extracted data and analytic code are provided in the Supplementary Materials. Additional materials are available from the corresponding author on reasonable request.

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